

DATA FORT

Personal Digital Vault

HOW TO RUN THE PROJECT

Final setup, database, configuration, credentials, and verification guide

This guide is the final replacement for the previous Data Fort run document. It adds the current package versions, the supplied SQL Server backup path, explicit secret-key configuration requirements, exact run commands, login credentials, and a verification checklist.

Edition	Final stable version documentation
Review date	14 September 2026
Backend target	.NET 10.0 / ASP.NET Core Web API
Frontend	Angular 21.2.x / TypeScript 5.9.x
Database	Microsoft SQL Server / EF Core 10.0.11
Local API	https://localhost:7166/api
Local UI	http://localhost:4200

Prepared from the supplied final stable ZIP and the previous Data Fort run guide. Version-sensitive details were reconciled against the actual project files and package-lock.json; no unimplemented BRD feature has been added.

1. What You Are Running

Data Fort is a full-stack Personal Digital Vault. The backend is an ASP.NET Core Web API targeting .NET 10.0. The frontend is an Angular 21 SPA. SQL Server stores application metadata and vault records; encrypted document bytes are stored locally by the API.

Component	Final local value
Solution	PersonalDigitalVault.sln
Backend project folder	PersonalDigitalVault.API
Frontend project folder	PersonalDigitalVault.Frontend
Database	PersonalDigitalVaultDB
Backend HTTPS	https://localhost:7166
Backend HTTP	http://localhost:5223 (redirected to HTTPS)
Swagger	https://localhost:7166/swagger (Development only)
Angular UI	http://localhost:4200
Frontend API base URL	https://localhost:7166/api

Final run recommendation

Use the HTTPS backend launch profile. It matches the API URL configured by the current frontend environment file and avoids a mismatch between Angular and the API.

2. Prerequisites

Prerequisite	Required / recommended
Operating system	Windows development environment is the intended setup in the supplied documentation.
.NET SDK	.NET 10 SDK; project target is net10.0.
Node.js	Node.js installed and available on PATH.
npm	The package metadata specifies npm@11.19.0; npm is used for package installation and Angular commands.
SQL Server	SQL Server, SQL Server Express, or LocalDB.
SSMS	Recommended for restore, inspection, and troubleshooting.
Browser	Chrome or Microsoft Edge recommended.

Check your tools

```
dotnet --version
node --version
npm --version
```

```
# Optional EF CLI check
dotnet ef --version
```

The final source does not include a global.json file, so the exact installed .NET 10 SDK patch version is not pinned by the repository. The application target and NuGet package versions are pinned in the project files.

3. Get the Project Ready

1. Extract the supplied ZIP so the solution file and both project folders remain in the same parent directory.
2. Open a terminal in the folder that contains **PersonalDigitalVault.sln**.
3. Do not depend on copied build output or copied node_modules. The correct workflow is restore/install from the project files.

4. Before public GitHub submission, remove the sensitive USER SECREATS.docx file from the repository and rotate any secrets that may already have been exposed.

The supplied final ZIP contains **USER SECREATS.docx**, which includes the runtime values for JWT, document encryption, and SMTP configuration. The values are intentionally not reproduced in this public-facing run guide. Use the file only as a secure local source, then store the settings through .NET User Secrets or another secure configuration provider.

Project folders

PersonalDigitalVault.API
PersonalDigitalVault.Frontend
PersonalDigitalVault.sln

4. Configure Backend Secrets

The API startup code requires the following settings. Non-secret values such as issuer, audience, and database connection are stored in `appsettings.json`; sensitive values should be supplied through User Secrets.

Key	Required for	Where to store
Jwt:Key	JWT signing and validation	User Secrets
DocumentEncryption:Key	AES encryption for credential passwords and document files	User Secrets
DocumentEncryption:KeyId	Identifies the active document encryption key	User Secrets
Smtp:Host	Email verification and password reset	User Secrets
Smtp:Port	SMTP connection port	User Secrets
Smtp:User	SMTP sender account	User Secrets
Smtp:Password	SMTP credential/app password	User Secrets

Recommended setup commands

```
cd PersonalDigitalVault.API
```

```
dotnet user-secrets init
dotnet user-secrets set "Jwt:Key" "<JWT_SECRET_FROM_SECURE_SOURCE>"
dotnet user-secrets set "DocumentEncryption:Key" "<BASE64_AES_KEY>"
dotnet user-secrets set "DocumentEncryption:KeyId" "<GUID>"
dotnet user-secrets set "Smtp:Host" "smtp.gmail.com"
dotnet user-secrets set "Smtp:Port" "587"
dotnet user-secrets set "Smtp:User" "<SMTP_ACCOUNT>"
dotnet user-secrets set "Smtp:Password" "<SMTP_APP_PASSWORD>"
```

The supplied `USER SECREATS.docx` is the secure source for the project-specific values. Do not put those real values into `appsettings.json`, frontend files, `README.md`, or public GitHub commits.

5. Database Setup - Recommended Backup Restore

The final stable ZIP contains **DIGITAL VAULT.bak**, a Microsoft SQL Server full database backup for PersonalDigitalVaultDB. This is useful when you need the closest reproduction of the submitted demo database state.

Recommended SSMS restore steps

- Open SQL Server Management Studio and connect to the SQL Server instance you will use locally.
- Right-click Databases -> Restore Database.
- Choose Device as the backup source and select DIGITAL VAULT.bak from the extracted project folder.
- Set the destination database name to PersonalDigitalVaultDB.
- Open the Files page and relocate the MDF/LDF physical paths if SSMS shows paths from another computer.
- Run the restore and confirm that PersonalDigitalVaultDB appears under Databases.

Important: the backup contains SQL Server file metadata tied to the machine on which it was created, so physical file paths may need to be changed during restore.

Alternative: create/update the schema with EF Core migrations

```
cd PersonalDigitalVault.API
dotnet restore
dotnet ef database update
```

The API startup initializer creates the User and Administrator roles if they do not exist. It does **not** seed the supplied test login accounts. Therefore, a completely fresh database can have the correct schema and roles but still not contain the two demo users.

6. Backend - Restore, Build, and Run

Open Terminal 1 and switch to the API project directory.

```
cd PersonalDigitalVault.API

dotnet restore
dotnet build
dotnet run --launch-profile https
```

Expected endpoints

Service	Address
API HTTPS	https://localhost:7166
API HTTP	http://localhost:5223
Swagger UI	https://localhost:7166/swagger

HTTPS certificate fix

```
dotnet dev-certs https --trust
```

Restart the API after trusting the development certificate if the browser still shows a certificate warning.

What happens at startup

- ASP.NET Core loads appsettings.json plus User Secrets in the Development environment.
- ApplicationDbContext connects to PersonalDigitalVaultDB.
- DbInitializer creates User and Administrator roles when needed.
- Swagger/OpenAPI is enabled because the Development environment is active.
- HTTPS redirection, CORS, JWT authentication, authorization, and API controller mapping are configured in the request pipeline.

7. Frontend - Install and Run

Open Terminal 2 and switch to the Angular project directory.

```
cd PersonalDigitalVault.Frontend
```

```
npm ci  
npm start
```

Expected frontend

<http://localhost:4200>

API target used by the current frontend

```
src/environments/environment.ts  
export const environment = {  
  production: false,  
  apiUrl: 'https://localhost:7166/api'  
};
```

Open the UI in Chrome/Edge and verify that the browser can reach the HTTPS API. The Angular auth interceptor adds the Bearer access token to authenticated requests after login.

Optional production-style build

```
npm run build
```

8. Login Credentials for Demonstration

The required test accounts provided for the Data Fort project are listed below.

Account	Email	Password	Role
Administrator	Piriyaram2004@gmail.com	Piriyaram2004@	Administrator
Normal User	Piriyaram2025@gmail.com	Piriyaram2025@	User

Expected login requirements

- Account must exist in the Users table.
- Account must be active (IsActive = true).
- Email must be verified (IsEmailVerified = true).
- Administrator account must reference the Administrator role; normal user must reference User role.
- The database restore route is preferred when you need the submitted demo state. A migration-only fresh database does not create these two users automatically.

Credential safety: these are project test credentials intended for evaluation. Do not reuse them as production credentials.

9. What to Verify After Startup

Check	Expected result
1. SQL Server	SQL Server instance is running and PersonalDigitalVaultDB is available.
2. Roles	User and Administrator roles exist.
3. API	https://localhost:7166 responds and API starts without secret/configuration exceptions.
4. Swagger	https://localhost:7166/swagger opens in Development.
5. Frontend	http://localhost:4200 loads the Angular app.
6. API base URL	Frontend calls https://localhost:7166/api .
7. Admin login	Administrator credentials authenticate and reach /admin pages.
8. User login	Normal User credentials authenticate and reach /vault pages.
9. Folder test	Create and edit a folder.
10. Document test	Upload, download, and verify integrity of a test document.
11. Credential test	Create and view a credential item.
12. Sharing test	Create a share link, open the public URL, then revoke it.
13. Admin test	View dashboard, users, activate/deactivate, and audit logs.
14. Email test	Registration verification and password-reset email flows work when SMTP is configured.

10. Troubleshooting

Problem	Fix
JWT key is not configured.	Set Jwt:Key using User Secrets and restart the API.
Document encryption key is not configured.	Set DocumentEncryption:Key; it must be a valid Base64-encoded AES key.
Document encryption key ID is not configured.	Set DocumentEncryption:KeyId to a valid GUID.
SMTP configuration is missing.	Set Smtp:Host, Smtp:Port, Smtp:User, and Smtp:Password in User Secrets.
Angular cannot call API.	Confirm the backend is running on https://localhost:7166 and the frontend environment points to https://localhost:7166/api.
HTTPS certificate warning.	Run dotnet dev-certs https --trust, then restart the browser/API.
Login says email is not verified.	The user record must have IsEmailVerified = true; restore the demo database or verify the account.
Login says invalid email/password.	Confirm the account exists, is active, and has the expected role/password hash. The startup initializer does not create the demo users.
EF command not found.	Install the EF CLI and retry: dotnet tool install --global dotnet-ef --version 10.0.11
Restore asks for missing MDF/LDF paths.	Use the SSMS Restore -> Files page to relocate the data/log files to paths available on your machine.
Port 7166 already in use.	Stop the conflicting local process or change the launch profile port and then update the Angular API base URL to match.

11. API and Feature Smoke Test

Use Swagger for API-level checks or the Angular UI for end-to-end checks. The stable API surface contains the following groups.

Group	Representative operations
Authentication	register, login, forgot-password, reset-password, change-password, profile, verify-email
Secure Vault	folders CRUD, documents CRUD/upload/download/integrity, credentials CRUD, search
Public Sharing	create/list/update/revoke/delete share links, public view/download by token
Administration	dashboard, users, user status, audit logs

Suggested evaluation path

- Login as Administrator and confirm administrator-only routes are visible.
- Log out and login as the Normal User.
- Create a folder and a nested subfolder.
- Upload a document, download it, and run integrity verification.
- Create an encrypted credential record.
- Create a share link, open the public route, and then revoke the link.
- Return to the Administrator account and verify audit/user dashboard views.

12. Final Submission Checklist

Item	Final check
GitHub repository	Repository is public; final stable source is pushed.
Technical Stack PDF	Upload Data_Fort_Full_Technical_Stack_Final.pdf.
How to Run PDF	Upload Data_Fort_How_to_Run_Project_Final_Updated.pdf.
Group name	Select the exact group name requested in the Google Form.
Credentials	Administrator and Normal User credentials are included in this guide.
Secrets	USER SECREATS.docx is not included in the public repository; rotate exposed secrets if it has already been committed publicly.
Database	Keep DIGITAL VAULT.bak as the local demo restore source; do not publish it unnecessarily.
Frontend/API match	Angular uses https://localhost:7166/api and backend runs on https://localhost:7166.

Final operational note

The final stable ZIP includes both EF Core migrations and a SQL Server backup. For a clean developer machine, either restore the supplied backup for the closest demo state or run the EF migrations to create/update the schema. When using migrations alone, the initializer creates roles but not the two demonstration accounts; use the restored demo database or otherwise ensure those accounts exist, are active, verified, and assigned to the correct roles.

End of How to Run the Project